

CV 5101

Modeling, Uncertainty, and Data for Engineers

(July – Nov 2026)

Dr. Prakash S Badal

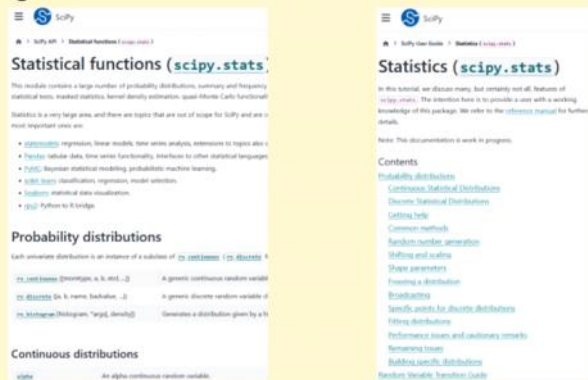
Flow

- Covariance
- Correlation
- Parametric distributions
- SciPy

L5
Sep 9, 2026

Announcement

- Practice `scipy.stats` for the following:
 - Using distributions → Shifting, scaling
 - Generating random numbers
 - Fitting distributions



scipy

- [Ref.] <https://docs.scipy.org/doc/scipy/reference/stats.html>
- Tutorial <https://docs.scipy.org/doc/scipy/tutorial/stats.html>
- Probability distributions
https://docs.scipy.org/doc/scipy/tutorial/stats/probability_distributions.html
- Run examples

- Common Methods
- Random number generator
- Shifting-scaling: loc, scale
- Fitting distributions

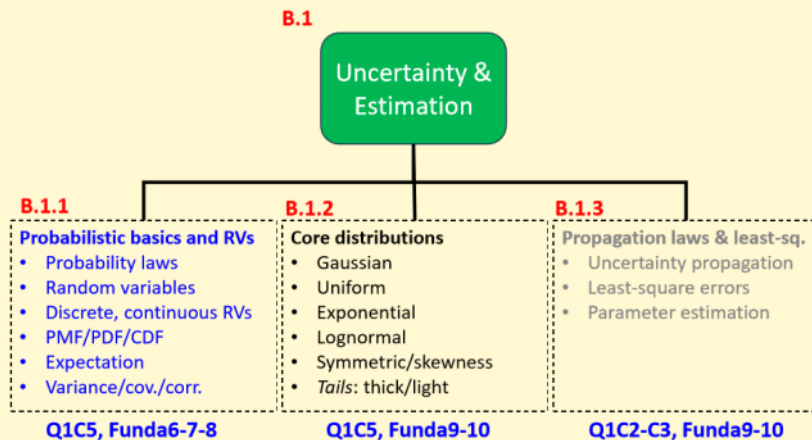
Check-in with teach-book

<https://mude.citg.tudelft.nl/book/2024/>

- Q1 Topics (Chapters 5, 6, 2, and 3)
 - **Q1C5 Univariate continuous distribution**
 - **Q1C5.1** PDF/CDF
 - **Q1C5.2** Empirical Distributions
 - **Q1C5.3** Parametric Distributions
 - Q1C5.4 Fitting a Distribution
 - Q1C6 Multivariate Distributions (briefly)
 - **Q1C2** Propagation of Uncertainty
 - Q1C3 Observation Theory: least-sq., Hyp. Test, Conf. Intervals
- Q2 Topics (Chapter 7 and 8)
 - Q2C7 Extreme value theory: GEV, return period, POT
 - Q2C8 Risk and decision making (CBA)
- Fundamental Concepts
 - **Chapter 6, 7, 8, 9.** Probability basics, rv, z- and t-tables
- Programming
 - Fundamental Concepts → Chapter 10

6

Module Overview



B.1.2 Core distributions (Q1C5)

Normal distribution: motivation

Human height

- Genetics
- Nutrition during 1000s of childhood
- Hormone production
- Sleep quality
- Skeleton response to mechanical stress

- Various additive effects
- Ubiquitous
- Mathematically tractable

Concrete strength

- Water-cement ratio
- Exact grain-size distribution
- Impurities
- Micro air-pockets

SAT score:

9

Normal (Gaussian) PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



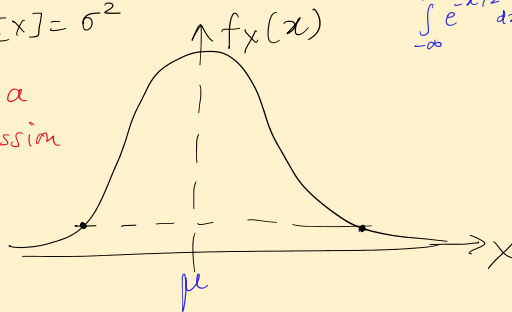
Gaussian/
Normal/
Bell-shaped

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$E[X] = \mu \quad ; \quad \text{Var}[X] = \sigma^2$$

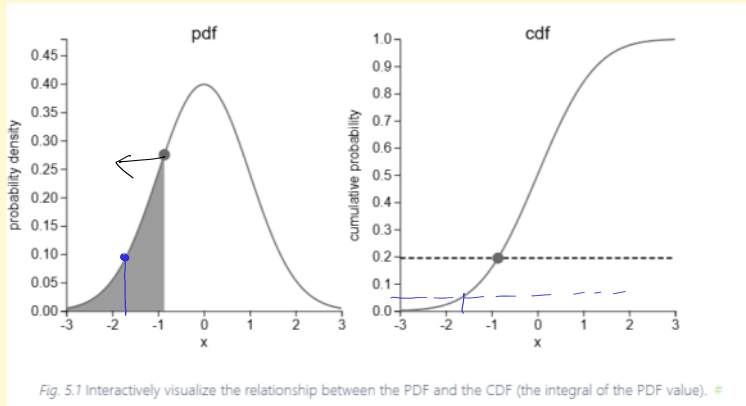
CDF does NOT have a
closed-form expression

$$\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$$



10

Gaussian



TeachBook Fig-5.1
Q1-CS

10

Standard Normal Probabilities

Z-table

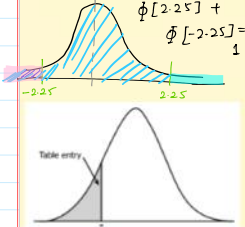
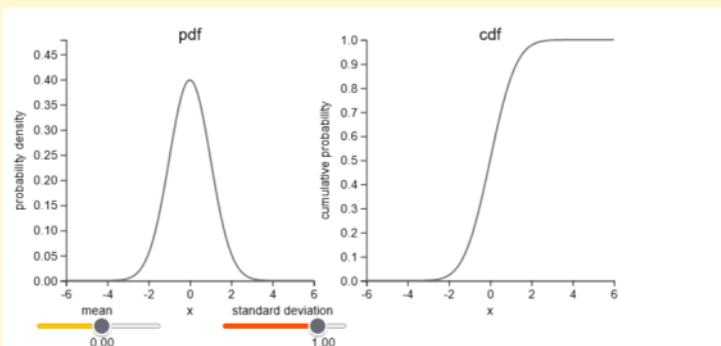


Table entry for z is the area under the standard normal curve to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

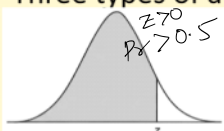
Gaussian



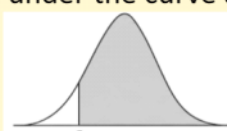
12

Normal distribution

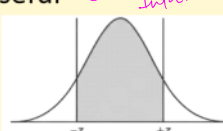
- Three types of areas under the curve are useful *Confidence Interval*



$\Pr(Z \leq z)$
 • Same as $\Phi(\cdot)$
 • Read this from table
 • Check if $>$ or $<$ 0.50

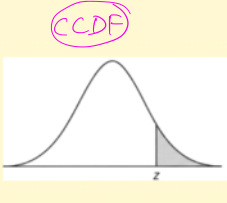
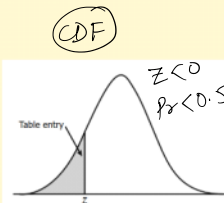


$\Pr(Z > z) = 1 - \Pr(Z \leq z)$
 • Same as $1 - \Phi(\cdot)$
 • Check if $<$ or $>$ 0.50



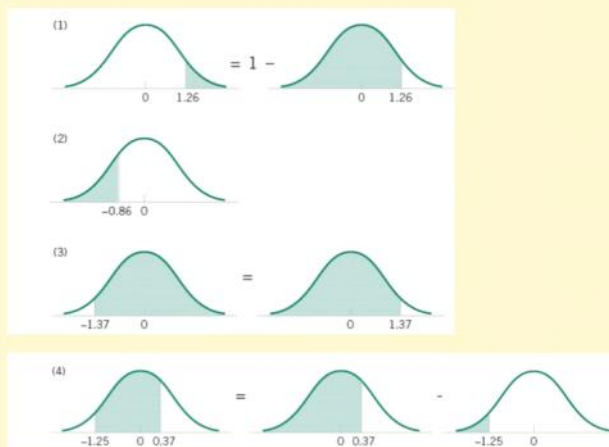
$\Pr(-z < Z < z)$
 $= \Pr(Z \leq z) - \Pr(Z \leq -z)$
 $= \Phi(z) - \Phi(-z)$
 $= 2\Phi(|z|) - 1$

• 68-95-99.7 rule
 $\begin{matrix} \longleftrightarrow & \longleftrightarrow & \longleftrightarrow \\ \pm 1\sigma & \pm 2\sigma & \pm 3\sigma \end{matrix}$



13

Examples (using basic prob. & symmetry)



14

Class problems

- Example 3:** A concrete batch has a mean strength of 10 MPa; variance 4 MPa²
- What is the probability that a test result exceeds 13 MPa?

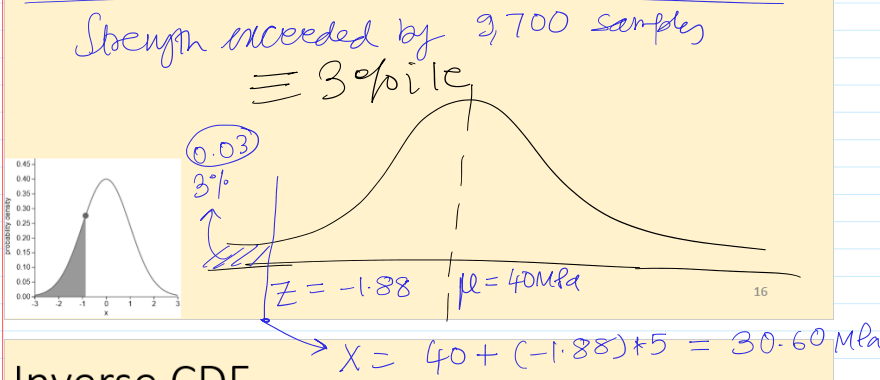
15

In-Class problem

- The compressive strength of a concrete mix is normally distributed with mean 40 MPa and standard deviation 5 MPa.

Out of 10,000 specimen produced, how many will generally have strength less than 35 MPa?

How many with strength less than 30 MPa?

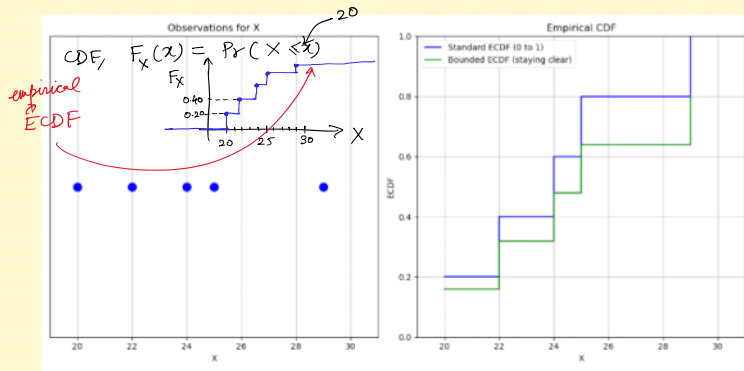


Inverse CDF

- For designing a structure, we often need a value that is not exceeded with more than p probability

$$x = F^{-1}(p)$$

Empirical distribution



Standard ECDF: $F_X(x) = \frac{i}{n}$, goes from 0 to 1.

Bounded ECDF: $F_X(x) = \frac{i}{n+1}$, stays clear of 0 & 1

18

Lx
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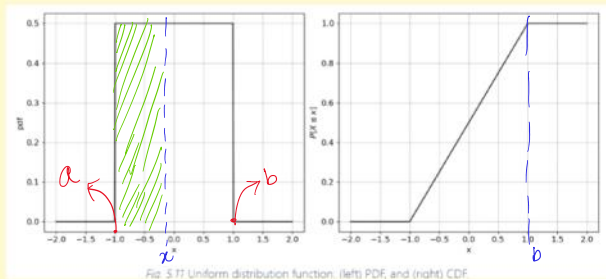
scipy

- Tutorial (not reference)
https://docs.scipy.org/doc/scipy/tutorial/stats/probability_distributions.html
- <https://docs.scipy.org/doc/scipy/tutorial/stats.html>
 - Run examples

- Common Methods
- Random number generator
- Shifting-scaling: loc, scale
- Fitting distributions

20

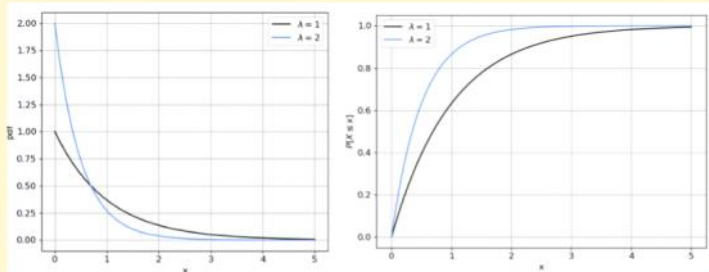
Uniform Distribution



PDF $f_X(x) = \begin{cases} \frac{1}{b-a} & ; a < x < b \\ 0 & ; \text{otherwise} \end{cases}$ **CDF** $F_X(x) = \begin{cases} 0 & ; x < a \\ \frac{x-a}{b-a} & ; a \leq x < b \\ 1 & ; x \geq b \end{cases}$

Properties $E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \frac{a+b}{2}$ $\text{Var}[X] = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f_X(x) dx = \frac{(b-a)^2}{12}$

Exponential Distribution



PDF $f_X(x) = \begin{cases} \lambda e^{-\lambda x} & ; x \geq 0, \lambda > 0 \\ 0 & ; \text{otherwise} \end{cases}$ **CDF** $F_X(x) = 1 - e^{-\lambda x} \quad x \geq 0$

Properties $E[X] = \frac{1}{\lambda} ; \text{Var}[X] = \frac{1}{\lambda^2}$

22

Lognormal Distribution: motivation

$X = X_1 + X_2 + \dots + X_n$
Additive effects $\rightarrow$ Norm-e Distribution

$X = X_1 \cdot X_2 \cdot \dots \cdot X_n$
Multiplicative effects

$\ln X = \ln X_1 + \ln X_2 + \dots + \ln X_n$
 $Y =$

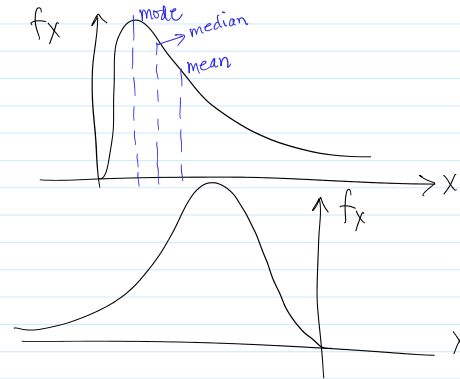
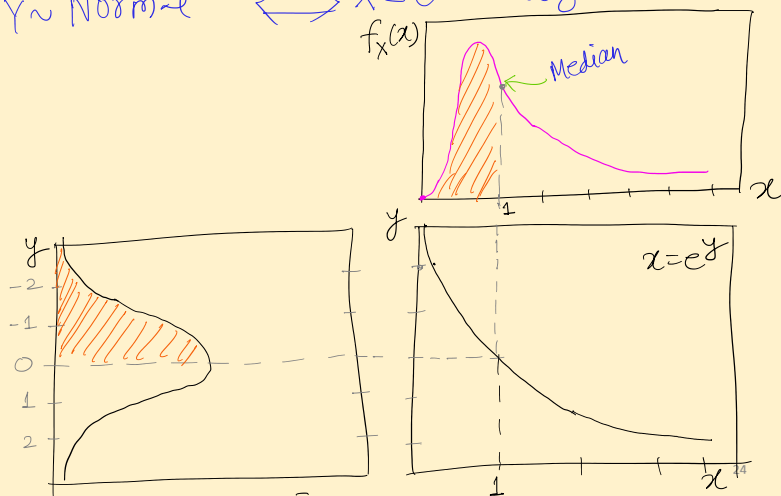
Multiplicative effects $\rightarrow$ Lognorm-e Distribution

23

Lognormal Distribution: transformation

$$X \sim \text{Lognormal} \iff Y = \ln X \sim \text{Normal}$$

$$Y \sim \text{Normal} \iff X = e^Y \sim \text{Lognormal}$$



Jacobian

• Let $Y = g(X)$

$f_X(x)$ is given.

Objective: obtain $f_Y(y)$

Assumption: X & Y have a one-to-one rel.

Equal probability:

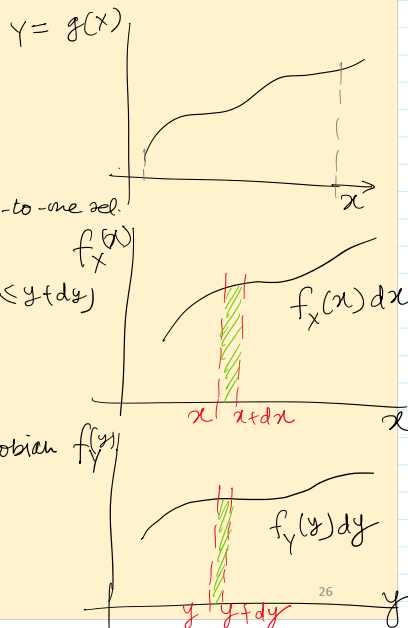
$$P_X(x < X \leq x+dx) = P_Y(y < Y \leq y+dy)$$

$$\Rightarrow f_X(x) dx = f_Y(y) dy$$

$$\Rightarrow f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| \rightarrow \text{Jacobian } f_Y(y)$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

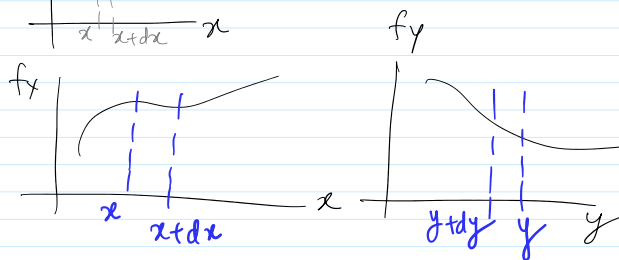
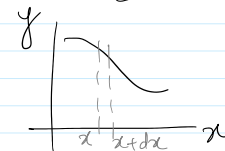
$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy$$



$$Y = [Y_1, Y_2]$$

$$X = [X_1, X_2]$$

$$J_{YX} = \begin{bmatrix} \partial y_1 / \partial x_1 & \partial y_1 / \partial x_2 \\ \partial y_2 / \partial x_1 & \partial y_2 / \partial x_2 \end{bmatrix}$$



Lognormal Distribution: parameters

• A random variable is Lognormal, if its log follows Normal
 $X \sim \text{Lognormal} \iff Y = \ln(X) \sim \text{Normal}$

Knowing $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, determine PDF of X

Recall, $\mathcal{N}(y; \mu, \sigma^2)$ $f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]$

Jacobian $f_X(x) = f_Y(y) \left| \frac{dy}{dx} \right|$ $dy/dx = 1/x$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma x} \exp\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right]$$

$$\lambda = \mu_Y = \mu_{\ln X} \quad \text{and} \quad \zeta = \sigma_Y = \sigma_{\ln X}$$

$$\mu_X = \exp\left(\lambda + \frac{\zeta^2}{2}\right)$$

$$\text{c.o.v. } \delta_X = \sqrt{\exp(\zeta^2) - 1}$$

Inverse rel.

$$\lambda = \ln \mu_X - \frac{\zeta^2}{2}$$

$$\zeta = \sqrt{\ln(1 + \delta_X^2)}$$

Gumbel distributions

- When we are interested in the smallest

or

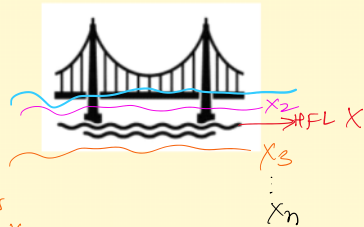
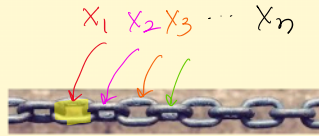
the largest of a set of RVs.

e.g., flood level under a bridge,
smallest link strength.

$$Y_1 = \min(X_1, X_2, \dots, X_n)$$

$$Y_n = \max(X_1, X_2, \dots, X_n)$$

Obj: knowing $X_1, X_2, \dots, X_n$ & their PDFs
what is the joint dist. of Y_1 & Y_n .



28

Gumbel distributions

$$Y_1 = \min(X_1, X_2, \dots, X_n),$$

$$Y_n = \max(X_1, X_2, \dots, X_n).$$

The CDF of Y_1 is (smallest extreme):

$$F_{Y_1}(y) = \Pr(Y_1 \leq y) = 1 - \Pr(Y_1 > y) = 1 - \prod_{i=1}^n \Pr(X_i > y)$$

$$\text{CDF } F_{Y_1}(y) = 1 - [1 - F_X(y)]^n$$

$$\text{PDF } f_{Y_1}(y) = n[1 - F_X(y)]^{n-1} f_X(y)$$

do at home!

29

Gumbel distributions

$$Y_1 = \min(X_1, X_2, \dots, X_n),$$

$$Y_n = \max(X_1, X_2, \dots, X_n).$$

The CDF of Y_n is (largest extreme):

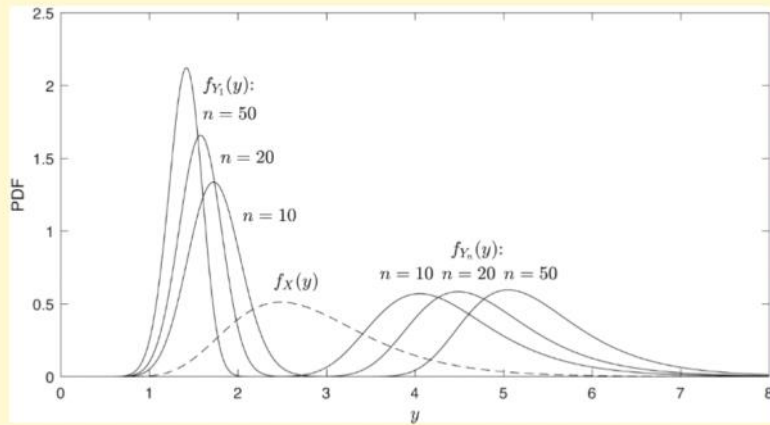
$$F_{Y_n}(y) = \Pr(Y_n \leq y) = \prod_{i=1}^n \Pr(X_i \leq y)$$

$$\text{CDF } F_{Y_n}(y) = [F_X(y)]^n$$

$$\text{PDF } f_{Y_n}(y) = n[F_X(y)]^{n-1} f_X(y)$$

30

Gumbel distributions



31

Gumbel distributions

Positive tail

PDF

CDF

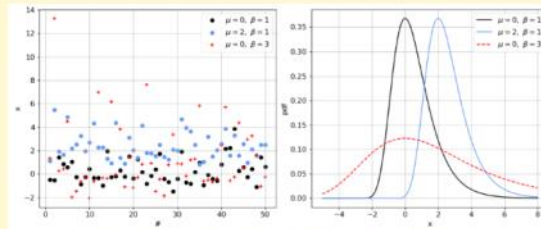


Fig. 5.14 Gumbel distribution function: (left) random samples, and (right) PDF. #

Negative tail

PDF

CDF

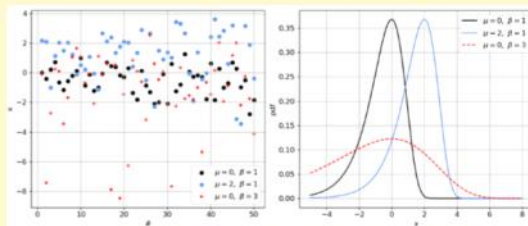
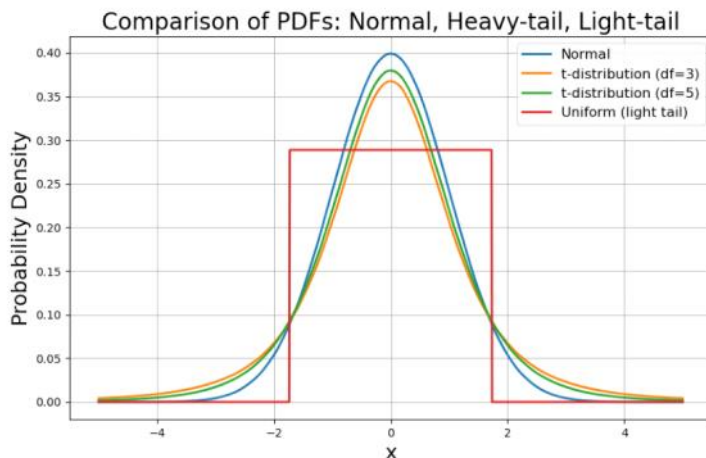


Fig. 5.16 Left-tailed Gumbel distribution function: (left) random samples, and (right) PDF. #

Core distributions

- Tails: thick/light



Next

B.1.3 Propagation laws &
least-squares
(Q1C3)

Questions, comments,
or concerns?